

Lecture 4 Probability Axioms

BIO210 Biostatistics

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LIFE SCIENCES

Probability theory is nothing but common sense reduced to calculation.

Laplace

Set

A set is a well-defined collection of distinct objects.

$$S = \{ \text{list or description of the objects in the set} \}$$

Sample space (Ω)

Set of all possible outcomes

Outcomes: **mutually exclusive** and **collectively exhaustive**

Sample space example 1

Example 1: flipping a coin four times

Sample space $\Omega = \{ \text{HHHH, HHHT, HHTH, HTHH,}$
 $\text{THHH, HHTT, HTHT, THHT,}$
 $\text{THTH, TTHH, HTTH, TTTH,}$
 $\text{TTHT, THTT, HTTT, TTTT} \}$

Sample space example 2

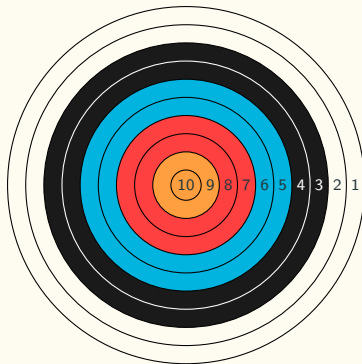
Example 2: an exam contained ten questions; each has 10 points; what is the total points you may get ?

Sample space $\Omega = \{0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$

Alternative sample space $\Omega = \{ 0, 10, 20, 30, 40, 50, 60, 70, 80, 90,$
100 and you are using your lucky pen,
100 and you are not using your lucky pen }

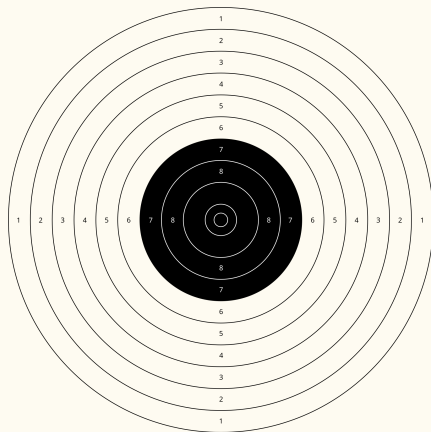
Sample space example 3

Example 3: shooting on a circular target (archery)



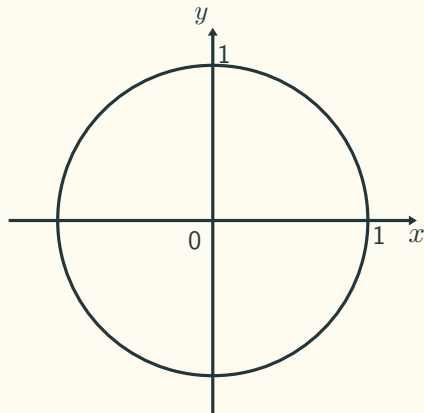
Sample space example 3

Example 3: shooting on a circular target (ISSF 10 Metre Air Pistol)



Sample space example 3

Example 3: positions on a target



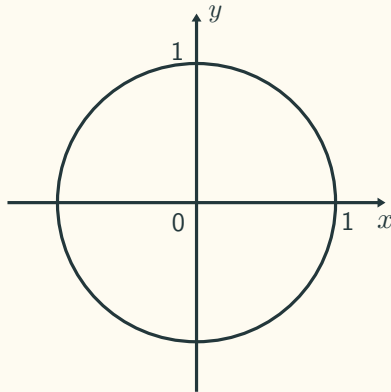
Sample space $\Omega = \{ (x, y) \mid x^2 + y^2 \leq 1 \}$

Assign probability to outcomes ... ?

Now, we can assign probability to individual outcomes ...

Not exactly!

What is the probability of hitting $(0, 0)$?

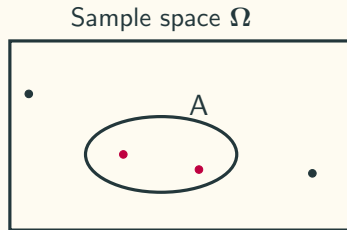


Event

Event

An event ($A, B, C, D, \text{etc.}$): a subset of the sample space Ω

- Probabilities are assigned to events. The probability represents our belief on how likely we think an event will occur.
- Event A has occurred. $\leftarrow$ what does this mean?



Probability axioms

FOUNDATIONS OF THE THEORY OF PROBABILITY

BY
A. N. KOLMOGOROV

TRANSLATION EDITED BY
NATHAN MORRISON

CHELSEA PUBLISHING COMPANY
NEW YORK

1950

§ 1. Axioms²

Let $\mathcal{E}$ be a collection of elements $\xi, \eta, \zeta, \dots$, which we shall call *elementary events*, and $\mathfrak{F}$ a set of subsets of E ; the elements of the set $\mathfrak{F}$ will be called *random events*.

- I. $\mathfrak{F}$ is a field³ of sets.
- II. $\mathfrak{F}$ contains the set E .
- III. To each set A in $\mathfrak{F}$ is assigned a non-negative real number $P(A)$. This number $P(A)$ is called the *probability of the event A* .
- IV. $P(E)$ equals 1.
- V. If A and B have no element in common, then

$$P(A + B) = P(A) + P(B)$$

A system of sets, $\mathfrak{F}$, together with a definite assignment of numbers $P(A)$, satisfying Axioms I-V, is called a *field of probability*.

The Kolmogorov Axioms

1. Nonnegativity: $\mathbb{P}(A) \geq 0$
2. Normalisation: $\mathbb{P}(\Omega) = 1$
3. Additivity: if A and B are disjoint ($A \cap B = \emptyset$), then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$

Nice properties

- The probability of any event is always between 0 and 1.
- If $A_1, A_2, A_3, \dots, A_n$ are disjoint, then

$$\mathbb{P}(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) + \dots + \mathbb{P}(A_n)$$

- $s_1, s_2, s_3, \dots, s_k$ are individual outcomes from the sample space, then

$$\begin{aligned}\mathbb{P}(\{s_1, s_2, s_3, \dots, s_k\}) &= \mathbb{P}(\{s_1\}) + \mathbb{P}(\{s_2\}) + \dots + \mathbb{P}(\{s_k\}) \\ &= \mathbb{P}(s_1) + \mathbb{P}(s_2) + \dots + \mathbb{P}(s_k) \leftarrow \text{abuse notation}\end{aligned}$$

Probabilities as long-term relative frequencies

If an experiment is repeated n times under essentially the identical conditions, and if the event A occurs m times, then as n grows large, the ratio $\frac{m}{n}$ approaches a fixed limit that is the probability of A :

$$\mathbb{P}(A) = \frac{m}{n}, \text{ where } n \text{ is large.}$$

Probabilities as a measure of belief

- The probability that my current manuscript gets published without revision is 1%.
- The probability that you will get a full score in BIO210 is 5%.
- The probability that it rains tomorrow is 80%.

Assigning probability

Experiment 1: flipping a fair coin four times

Sample space $\Omega = \{ \text{HHHH, HHHT, HHTH, HTHH,}$
 $\text{THHH, HHTT, HTHT, THHT,}$
 $\text{THTH, TTHH, HTTH, TTTH,}$
 $\text{TTHT, THTT, HTTT, TTTT} \}$

All possible outcomes are **equally likely**, so we can let every possible outcome have a probability of $1/16$.

Calculate the probabilities of the following events:

$A = \{\text{all heads or tails}\}$

$B = \{\text{exactly two head}\}$

$C = \{\text{at least two tails}\}$

Discrete Uniform Law

Let all outcomes be equally likely, then

$$\mathbb{P}(A) = \frac{\text{number of elements of } A}{\text{total number of sample points}} = \frac{|A|}{|\Omega|}$$

Computing probability is essentially just counting!

Continuous uniform law

Experiment 2: archery

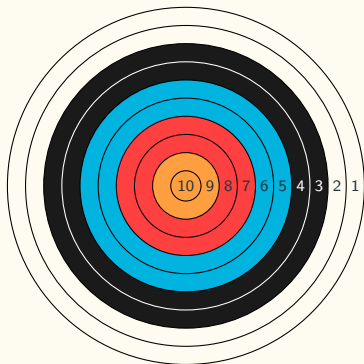
Sample space $\Omega = \{ (x, y) \mid x^2 + y^2 \leq 1 \}$

All possible outcomes are equally likely, Then probability = the ratio of areas.

$A = \{\text{hitting the red area}\}$

$B = \{ (x, y) \mid x + y \leq 1 \}$

$C = \{(0, 1), (1, 0), (0, -1), (-1, 0)\}$



Experiment 3: keep flipping a fair coin until you obtain a head for the first time and stop.

Sample space $\Omega = \{ H, TH, TTH, TTTH, TTTTH, \dots \}$

Let n be the number of flips, $\mathbb{P}(n) = \frac{1}{2^n}$, $n = 1, 2, 3, 4, \dots$

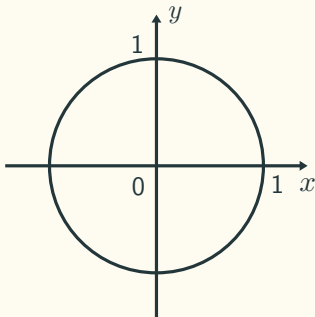
$A = \{ n \text{ is an even number} \}$, $\mathbb{P}(A) = ?$

Countable Additivity Axiom

If a sequence of events $A_1, A_2, A_3, \dots$ are disjoint, then

$$\mathbb{P}(A_1 \cup A_2 \cup A_3 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) + \dots$$

Misuse of the countable additivity axiom



Sample space $\Omega = \{ (x, y) \mid x^2 + y^2 \leq 1 \}$

Paradox 1??

$$1 = \mathbb{P}(\Omega) = \mathbb{P}\left(\bigcup \{(x, y)\}\right) = \sum_{x, y} \mathbb{P}(\{(x, y)\}) = \sum_{x, y} 0 = 0$$

Take-home message: $\{(x, y)\}$ is uncountable: it is not possible to list every single one of (x, y) .

Paradox 2??

An experiment is performed, and the outcome is $\left(\frac{1}{2}, \frac{1}{2}\right)$

Take-home message: probability of 0 does NOT mean impossible.